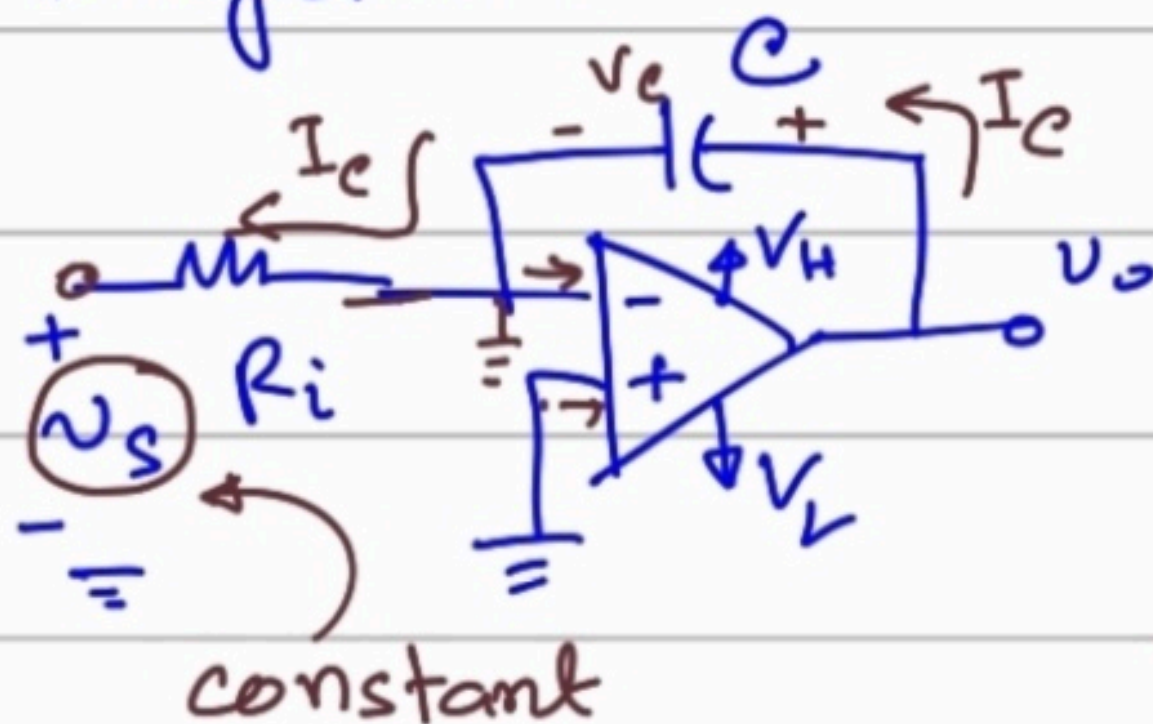


Triangular Wave Generator:

Integrator:



We are assuming virtual ground is present here.

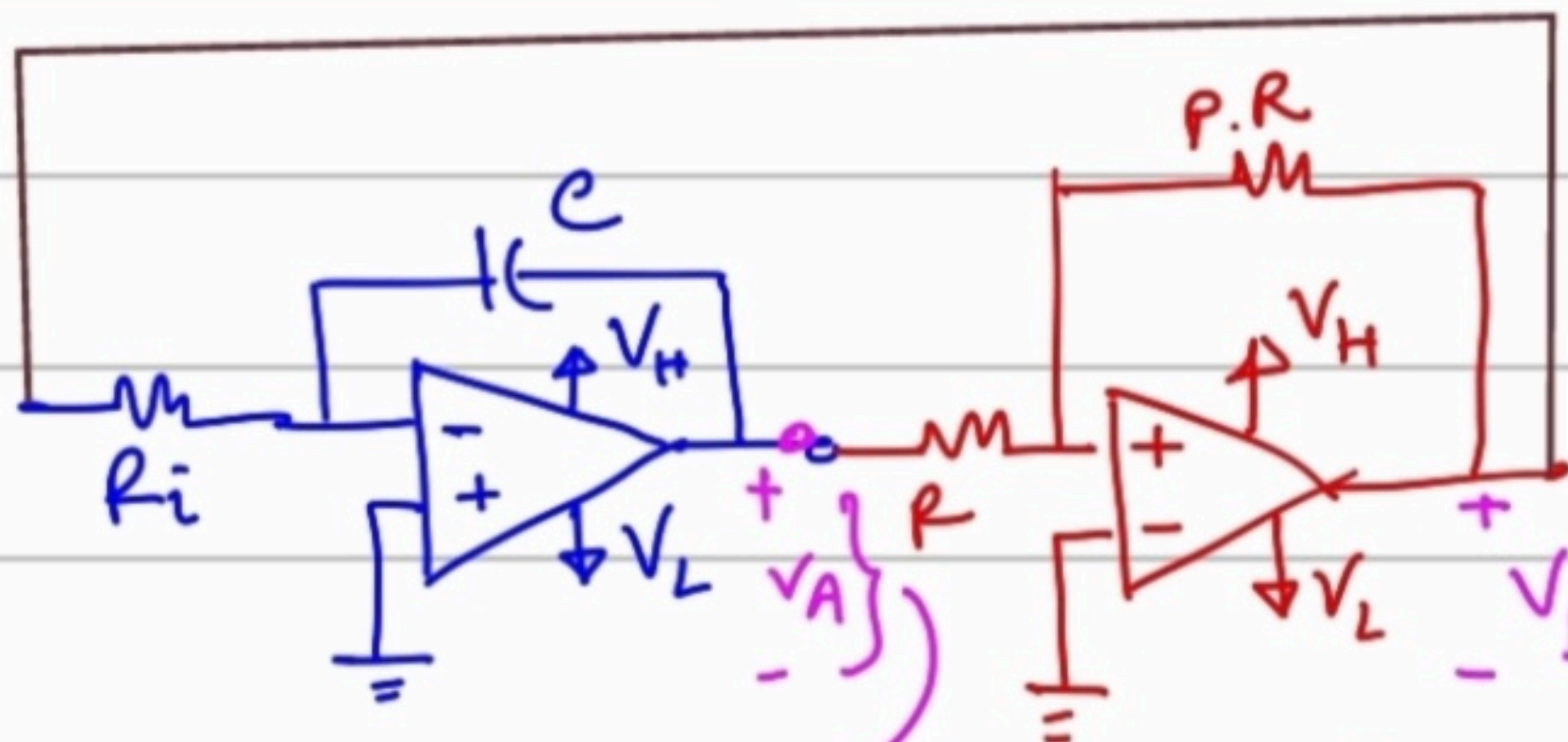
$$I_c = C \frac{dV_c}{dt} = \frac{0 - V_s}{R_i}$$

$$\Rightarrow \int_{V_c(t_1)}^{V_c(t)} dV_c = - \frac{V_s}{R_i C} \int_{t_1}^t dt$$

$$\Rightarrow V_c(t) - V_c(t_1) = - \frac{V_s}{R_i C} (t - t_1)$$

$$\Rightarrow V_c(t) = V_c(t_1) - \frac{V_s}{R_i C} (t - t_1)$$

$$V_c(t_1) = V_{initial}$$



Integrator

Non-inverting Schmitt trigger

Triangular wave

Square wave

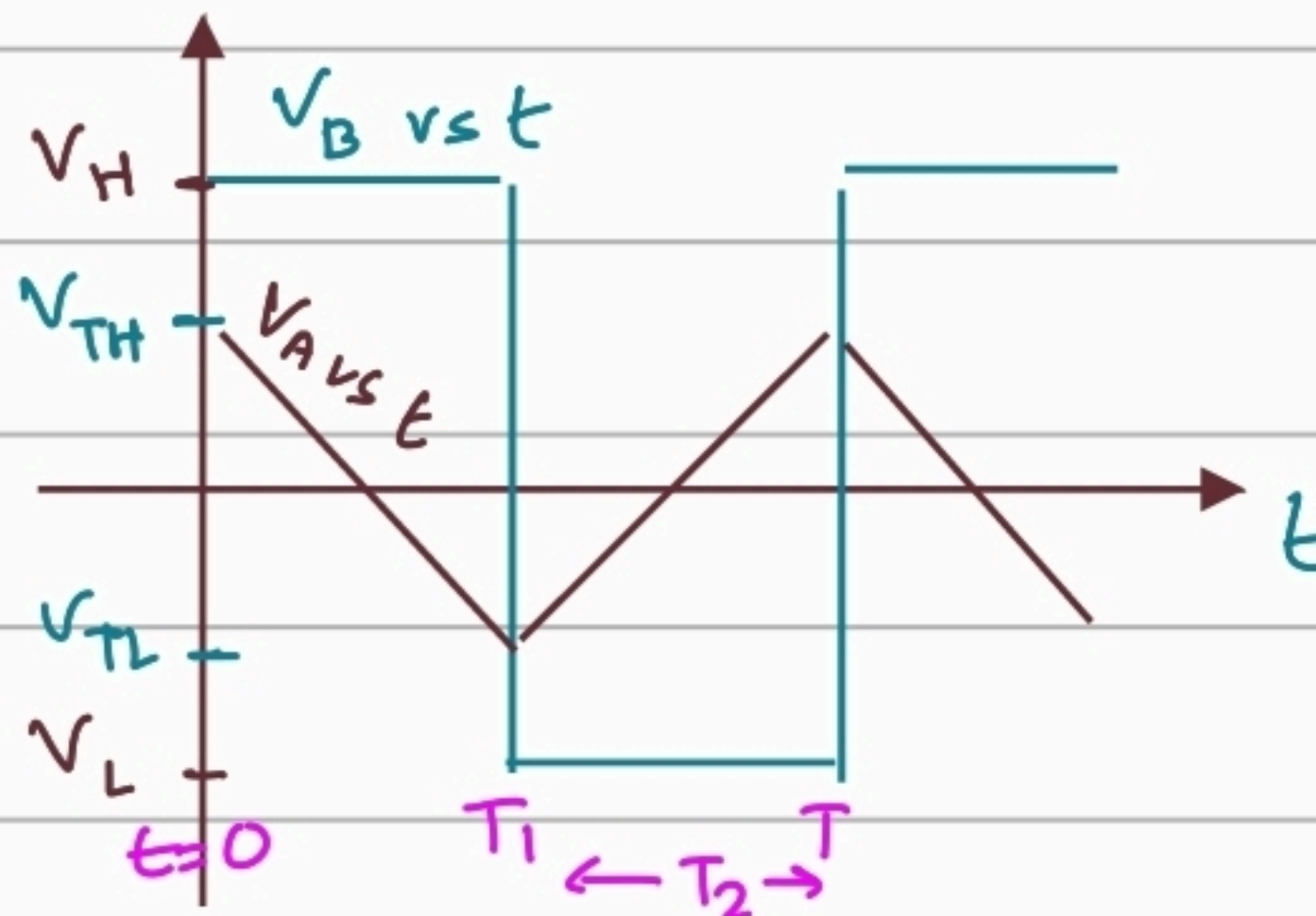
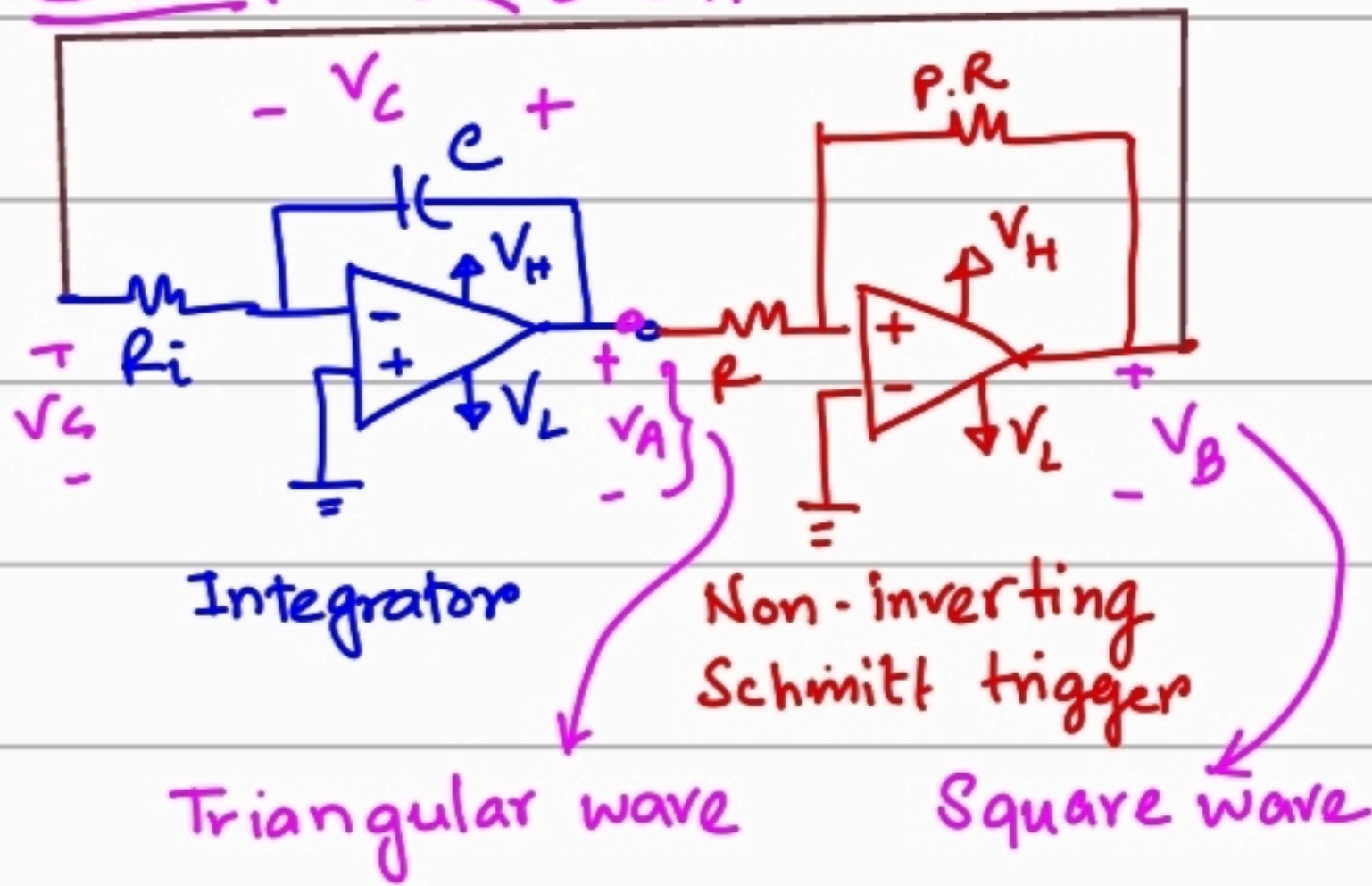
$$V_{TL} = V_H \left(- \frac{R}{PR} \right)$$

$$V_{TL} = - \frac{V_H}{P}$$

$$V_{TH} = - \frac{V_L}{P}$$

$$\begin{cases} V_{UT} = V_{TH} \rightarrow \text{upper or higher Threshold} \\ V_{LT} = V_{TL} \rightarrow \text{lower Threshold} \end{cases}$$

Case: 1 : $0 < t < T_1$



Initial condition: $v_B = v_H$ and $v_A = v_{TH}$, $t_{initial} = 0$

$$V_C(t) = V_A(t) = V_{\text{initial}} - \frac{V_s}{R_i C} (t - t_{\text{initial}}) \quad [V_s = V_B = V_H]$$

$$\rightarrow V_A(t) = V_{TH} - \frac{V_H}{R_i C} (t - 0)$$

Because $V_A(T_1) = V_{TL}$, we can find.

$$T_1 = R_i C \left(\frac{V_{TH} - V_{TL}}{V_H} \right)$$

Case 2: $T_1 < t < T$. Using similar analysis like square wave we can derive,

$$T_2 = R_i C \left(\frac{V_{TL} - V_{TH}}{V_L} \right) \quad H \leftrightarrow L$$

Total time period, $T = T_1 + T_2$. frequency, $f = \frac{1}{T}$

Special case: $V_L = -V_H$ $V_{TH} = -\frac{V_L}{p}$, $V_{TL} = -\frac{V_H}{p}$

$$f = \frac{\phi}{4R_i c}$$

Unipolar Triangular Wave Generator